

Concept Proposal

Module: SE4051 - Trends in Digital Media

Project Title: The Sentient Forest: Flowbound Grove

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Title of the Environment

The Sentient Forest: Flowbound Grove

Scenario / Application Domain

Flowbound Grove is an intelligent and interactive digital environment developed in Unity. The environment simulates a living forest where the player explores, collects Light Seeds, and restores the Forest Heart. The project is designed as a responsive digital media experience that combines gameplay, adaptive system behavior, and user-centered interaction.

Target User

The primary target users are students and young adults (approximately ages 16 to 30) who engage with interactive exploration experiences. The design also supports users who require guided navigation or lower-intensity interactions through built-in accessibility controls.

How the System Demonstrates Intelligent Behavior

The system demonstrates intelligent behavior through multiple rule-based and context-aware interactions:

- Light Seeds detect player proximity and trigger dynamic collection events.
- Shadow Wisps detect distance to the player and shift from passive to follow behavior.
- Luma (guide NPC) adapts warnings and guidance according to player stress and location.
- Forest Heart validates completion conditions and unlocks the restoration sequence only when required objectives are completed.
- GameManager coordinates objectives, stress updates, system feedback, and progression states.

Planned Adaptive Features

- Dynamic fog and lighting changes in Shadow Zones.
- Context-based audio transitions (calm, danger, collect, and win states).
- Stress-based system adaptation with continuous real-time updates.
- Assistive guide path activation at high stress levels or manual request.
- Reduced-intensity accessibility mode for improved usability.
- Responsive user interface across different resolutions and aspect ratios.

Intended User Experience

The intended experience is immersive, clear, and supportive. The user should feel that the world is aware of their actions and condition. Through immediate visual, audio, and interface feedback, the system guides users from initial exploration to final restoration. The design balances challenge and assistance by providing adaptive support during stressful moments while preserving player agency and engagement.